



DPUSGEDistortionObject

Distorts the image of a SGEView and displays it on a custom object.

1. Static parameters:

<i>ViewID</i>	The number of the view whose image is distorted. Counting starts at 0, which is assigned to the first view that is listed in the configuration.
<i>ResolutionX</i> , <i>ResolutionY</i>	Size of the off-screen image buffer (in pixels), into which the scenery is rendered. Should correspond to the image size after distortion. Large Resolution values reduce the achievable frame rate. Note: For older graphics boards (e.g. NVidia GeForce 6000, 7000 series), powers of two must be used for ResolutionX and ResolutionY, otherwise no output can be seen. Newer boards can use arbitrary values.
<i>Object</i>	Resource name of the object to load. The object must be available in the SIG or SGE file format. The object model must be located in a subfolder of SI-LAB\lib\graphics\default\models. For more details, please see the documentation of DPUSGEObject.
<i>DistortResetViewport</i>	Selects whether the 2D coordinates given in the object file are relative to the whole window into which the view is rendered (1) or only to the selected viewport (0). The default value is 0 and only needs to be changed if the object's shape is very complex.

2. Dependencies to other DPUs:

DPUSGEDistortionObject must run together with **DPUSGEWorld** and **DPUSGEView**.

The following Index values must be assigned:

`DPUSGEWorld.Index < DPUSGEDistortionObject.Index < DPUSGEView.Index`

3. Functional principle:

The DPU works well with the following script:

```
DPUSGEDistortionObject DistortMirror
{
    Computer = {VISM};
    Index = 50;
    ViewID = 1;
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
// This object file defines the size and shape of the displayed view
Object = mirrors.mirror_left;

// Internal resolution in which the scenery is rendered
ResolutionX = 1120;
ResolutionY = 960;

};
```

The script assumes that mirrors are (using *DPUSGEGView SGEMirror*) inserted on top of a basic front-screen view (which is generated by *DPUSGEGView SGEView*).

Using the object, an arbitrary geometry of the mirror can be defined, which is not limited by a mathematical formula (like when using *DPUSGEDistortionBezier*).

The object will usually be specified in screen coordinates, and it will be placed relative to the viewport of *SGEMirror*.

The purpose of the object is to map the scenery texture (S and T coordinates from 0..1) to arbitrary geometry in screen coordinates (X and Y coordinates from 0..*SizeInPixels*).

A simple example of a suitable object follows:

```
define SIG3DObject Mirr
{
    Animation = default;
    Type = (5,0,0);
    AlphaTest = 0.2;
    Blend = true;
    Color = (1,1,1,1);
    RenderClass = Overlay;

    Vertices =
    {
        // upper section
        // Texture   Pixel coord.
        // S  T      X    Y    Z
        ([0.75,0.7],< 0.0, 0.0,0.0>),
        ([0.75,0.3],< 0.0, 382.0,0.0>),
        ([0.95,0.3],<224.0,382.0,0.0>),
        ([0.95,0.7],<224.0,0.0,0.0>),

        // lower section
        // Texture   Pixel coord.
        // S  T      X    Y    Z
        ([0.0,1.0],< 0.0, 384.0,0.0>),
        ([0.0,0.0],< 0.0, 576.0,0.0>),
        ([1.0,0.0],<224.0,576.0,0.0>),
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
        ([1.0,1.0],<224.0,384.0,0.0>),  
    };  
  
    Primitives[0] =  
    {  
        Quads  
        {  
            0,1,2,3,  
            4,5,6,7  
        };  
    };  
};
```

This object maps the whole scenery to a smaller rectangle in the lower area (“wideangle mirror”). Additionally, a smaller, central part of the scenery is mapped to a larger area on top (“narrow mirror”).